



Addressing Distribution Shift for Robust and Trustworthy Prediction and Causal Inference in Clinical AI Settings

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Distribution shift, or data shift, is a fundamental challenge in generalizing or transporting scientific evidence from observed source populations to new, potentially only partially observed target populations. In medical research, data from external or historical randomized clinical trials or observational studies are often used to estimate treatment effects for a new target population. Subasri et al¹ empirically examined the deployment of state-of-the-art machine learning (ML) and artificial intelligence (AI) algorithms across clinical datasets under distribution shifts. Analyzing a retrospective cohort of 109 802 patients across 7 hospitals in Toronto, Canada (2010-2020), spanning both academic and community settings, they evaluated ML and AI predictions of 14-day all-cause in-hospital mortality using electronic health record data. Their work highlights both the potential and challenges of applying ML and AI algorithms in heterogeneous clinical environments, emphasizing the need for robust strategies to handle distribution shifts.

To detect and mitigate harmful shifts affecting model performance, Subasri et al¹ developed a proactive, label-agnostic monitoring pipeline, combining dimensionality reduction and statistical testing. Their approach identified shifts linked to hospital type (eg, nursing homes vs acute care centers), patient demographics (eg, age <65 years and sex), and laboratory assays (eg, troponin, brain natriuretic peptide [BNP], and D-dimer). For example, shifts in BNP assays led to area under the receiver operating characteristic curve (AUROC) declines of 0.23 for neurodevelopmental disorders and 0.26 for nervous system disorders. To mitigate performance degradation, the authors used transfer learning, finding that multihospital training improved accuracy, particularly in community hospitals (AUROC or area under the precision-recall curve gains of 0.05-0.06). Selective retraining proved less effective than continuous dataset updates, highlighting the importance of proactive monitoring and broad data integration.

Addressing distribution shift has become a major focus in statistics and ML and AI, driving extensive methodologic progress. This commentary highlights 3 key areas of recent progress and ongoing challenges: (1) moving beyond covariate shift to a broader understanding of distribution shift, (2) moving beyond prediction tasks toward causal inference, and (3) moving beyond classical asymptotics to distribution-free, finite-sample uncertainty quantification for black-box ML and AI predictions. Connecting these advances with clinical applications requires collaborative innovation to ensure that methodologic rigor translates into improved health care outcomes.

Many statistical methods assume that observed covariates account for all distributional differences between populations. Covariate shift, a key assumption, posits that while covariate distributions may differ across populations, the conditional outcome distribution remains unchanged given the covariates. For example, age distributions might vary, but treatment effects are assumed homogeneous for individuals with the same covariate profile. Although reweighting based on density ratios can adjust for covariate shifts, recent research suggests it is often insufficient. Empirical evaluations of generalization and transportation estimators show that observed covariate shift explains only a small fraction of clinical distribution shifts.² This finding implies that adjusting for observed covariate shift alone may be insufficient for generalization and transportation and that the remaining unobserved conditional shift—that is, the shift in the conditional distribution of outcomes given the covariates—is often larger. Existing methods either assume no shift in the conditional distribution, which may be unrealistic, or assume a bounded shift and adopt worst-case scenario

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analyses, which tend to be overly adversarial.² A 2024 study by Jin et al³ provides a data-adaptive alternative, showing that unobserved conditional shift can often be predicted from observed covariate shift. In the future, more nuanced sensitivity analyses and adaptive methods are needed to move beyond covariate shift assumptions to offering more effective strategies.

Generalizing or transporting causal inferences across populations, settings, or time points is even more challenging than in prediction tasks. Traditional inverse probability weighting methods often suffer from instability and model misspecification, prompting the development of doubly robust and balancing weights approaches that yield more stable, efficient, and reliable effect estimates. A 2023 comprehensive review by Brantner et al⁴ highlights substantial progress in addressing these challenges. Federated or multisite studies are particularly valuable for estimating treatment effects in underrepresented populations, offering increased statistical power to study rare exposures and outcomes. However, these studies face challenges due to privacy constraints that prevent the sharing of individual-level data and distributional shifts across sites. Recent advances in federated causal inference mitigate these challenges through data-adaptive methods that upweight informative sites to improve efficiency while downweighting noninformative sites to reduce bias. A penalized regression framework ensures the safe incorporation of source sites while preventing negative transfer.⁵ A key consideration in developing federated learning methods is ensuring they are communication efficient, allowing sites to exchange summary statistics only once. Ongoing advancements in federated causal survival analysis, semisupervised learning, and prediction-powered or surrogate-assisted estimation of long-term treatment effects hold particular promise. As multicenter research networks become increasingly common in clinical research, federated approaches will play an essential role in producing robust and generalizable causal inferences.

Finally, conformal inference is an exciting and rapidly evolving area of modern statistical methodology that is particularly well-suited for quantifying uncertainty in predictions from black-box ML and AI models in multienvironment settings. These methods provide distribution-free, finite-sample valid prediction sets or intervals, ensuring reliable uncertainty quantification regardless of the underlying data distribution. Originally developed under the assumption of exchangeability between test and calibration data, conformal prediction has since been extended to handle covariate shift. Recent methodologic advancements have further expanded its applicability. Yang et al⁶ established connections between conformal prediction, missing data, and semiparametric efficiency theory, broadening its theoretical foundation. In multisource data settings, Liu et al⁷ introduced methods that explicitly account for both covariate shift and conditional shift, addressing key limitations of existing approaches. Conformal inference under distribution shift continues to advance rapidly, extending into novel settings such as multivariate outcomes, time-to-event data, time-series analysis, and online learning. These developments highlight its growing importance in ensuring robust, generalizable inference across heterogeneous data environments.

Despite significant methodologic progress, practical adoption of clinical ML and AI models that accommodate distribution shift lags behind. Although Subasri et al¹ demonstrate successful deployment across multisource clinical environments, clinical uptake remains limited. Key barriers include lack of easily implementable software, accessible code, clear tutorials, and educational resources. Additionally, many critical clinical challenges remain underexplored. For example, conformal prediction for multivariate outcomes could enable simultaneous evaluation of efficacy and safety outcomes, accelerating regulatory approvals, especially crucial in public health crises such as pandemics. Bridging these gaps requires stronger collaboration between statisticians, ML and AI researchers, software engineers, clinicians, policy makers, and patients. Ultimately, ensuring that methodologic advances reach the front lines of clinical care is essential for realizing their full potential to improve patient outcomes.

ARTICLE INFORMATION

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REFERENCES

1. Subasri V, Krishnan A, Kore A, et al. Diagnosing and remediating harmful data shifts for the responsible deployment of clinical AI models. *JAMA Netw Open*. 2025;8(6):e2513685. doi:[10.1001/jamanetworkopen.2025.13685](https://doi.org/10.1001/jamanetworkopen.2025.13685)
2. Mu T, Chandak Y, Hashimoto TB, Brunskill E. Factored DRO: factored distributionally robust policies for contextual bandits. In: Koyejo S, Mohamed S, Agarwal A, Belgrave D, Cho K, Oh A, eds. *NIPS'22: 36th International Conference on Neural Information Processing Systems*. Curran Associates Inc; 2022:8318-8331.
3. Jin Y, Egami N, Rothenhäusler D. Beyond reweighting: on the predictive role of covariate shift in effect generalization. *arXiv*. Preprint posted online December 12, 2024. doi:[10.48550/arXiv.2412.08869](https://doi.org/10.48550/arXiv.2412.08869)
4. Brantner CL, Chang TH, Nguyen TQ, Hong H, Stefano LD, Stuart EA. Methods for integrating trials and non-experimental data to examine treatment effect heterogeneity. *Stat Sci*. 2023;38(4):640-654. doi:[10.1214/23-STS890](https://doi.org/10.1214/23-STS890)
5. Han L, Hou J, Cho K, Duan R, Cai T. Federated adaptive causal estimation (face) of target treatment effects. *J Am Stat Assoc*. Published online March 17, 2025. doi:[10.1080/01621459.2025.2453249](https://doi.org/10.1080/01621459.2025.2453249)
6. Yang Y, Kuchibhotla AK, Tchetgen Tchetgen E. Doubly robust calibration of prediction sets under covariate shift. *J R Stat Soc Series B Stat Methodol*. 2024;86(4):943-965. doi:[10.1093/jrsssb/qkae009](https://doi.org/10.1093/jrsssb/qkae009)
7. Liu Y, Levis AW, Normand SL, Han L. Multi-source conformal inference under distribution shift. *Proc Mach Learn Res*. 2024;235:31344-31382.